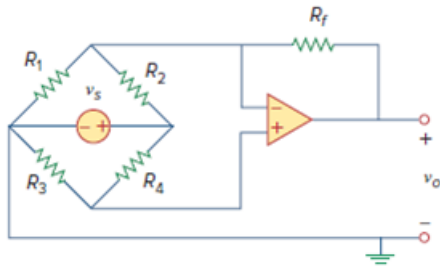


Problem

In the circuit shown in Fig. 5.62, find k in the voltage transfer function $v_o = k v_s$.

Figure 5.62:



Step-by-step solution

Step 1 of 4

Refer to circuit diagram in Figure 5.62 in the textbook.

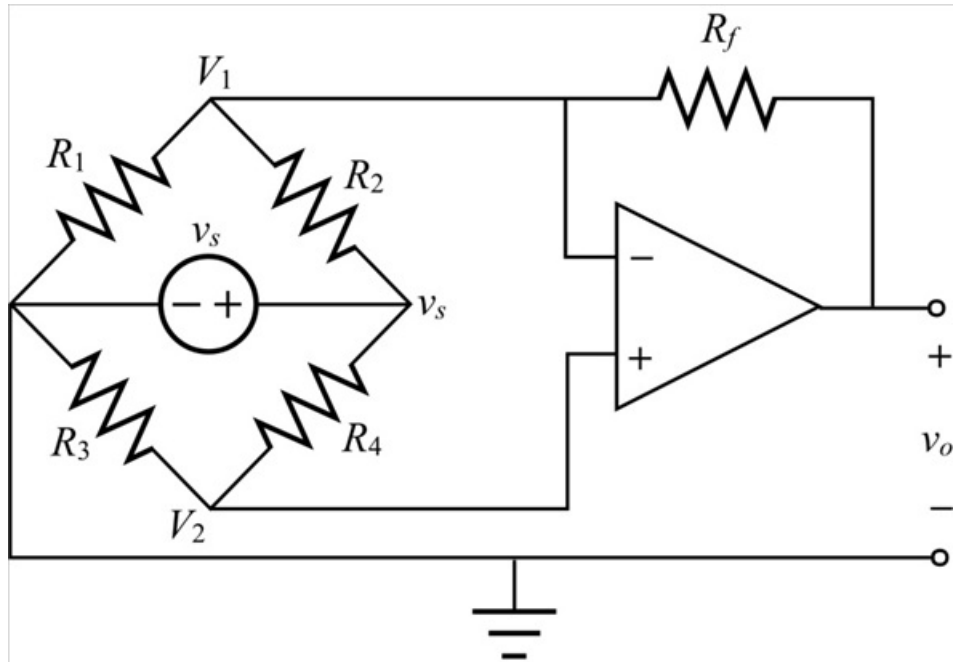


Figure 1

Step 2 of 4

Apply Kirchhoff's current law at node V_1 .

$$\frac{V_1}{R_1} + \frac{V_1 - v_s}{R_2} + \frac{V_1 - v_o}{R_f} = 0$$

$$\frac{V_1}{R_1} + \frac{V_1}{R_2} + \frac{V_1}{R_f} = \frac{v_s}{R_2} + \frac{v_o}{R_f}$$

$$\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_f} \right) V_1 = \frac{v_s}{R_2} + \frac{v_o}{R_f} \dots\dots (1)$$

Apply Kirchhoff's current law at node V_2 .

$$\frac{V_2}{R_3} + \frac{V_2 - v_s}{R_4} = 0$$

$$\frac{V_2}{R_3} + \frac{V_2}{R_4} = \frac{v_s}{R_4}$$

$$\left(\frac{R_3 + R_4}{R_3 R_4} \right) V_2 = \frac{v_s}{R_4}$$

$$V_2 = \frac{R_3 v_s}{R_3 + R_4} \dots\dots (2)$$

Step 3 of 4

According to virtual ground concept, the voltages at inverting and non-inverting terminals of op-amp are same.

Therefore,

$$V_1 = V_2$$

Recall equation (1).

$$\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_f} \right) V_1 = \frac{v_s}{R_2} + \frac{v_o}{R_f}$$

Substitute V_2 for V_1 .

$$\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_f} \right) V_2 = \frac{v_s}{R_2} + \frac{v_o}{R_f}$$

Substitute $\frac{R_3 v_s}{R_3 + R_4}$ for V_2 .

$$\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_f} \right) \frac{R_3 v_s}{R_3 + R_4} = \frac{v_s}{R_2} + \frac{v_o}{R_f}$$

$$\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_f} \right) \left(\frac{R_3}{R_3 + R_4} \right) v_s - \frac{v_s}{R_2} = \frac{v_o}{R_f}$$

$$\left[\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_f} \right) \left(\frac{R_3}{R_3 + R_4} \right) - \frac{1}{R_2} \right] v_s = \frac{v_o}{R_f}$$

$$v_o = R_f \left[\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_f} \right) \left(\frac{R_3}{R_3 + R_4} \right) - \frac{1}{R_2} \right] v_s \dots\dots (3)$$

Step 4 of 4

The transfer function of the circuit is,

$$v_o = kv_s \dots\dots (4)$$

Equate equation (3) and equation (4).

$$k = R_f \left[\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_f} \right) \left(\frac{R_3}{R_3 + R_4} \right) - \frac{1}{R_2} \right]$$

Therefore, the value of k is $R_f \left[\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_f} \right) \left(\frac{R_3}{R_3 + R_4} \right) - \frac{1}{R_2} \right]$.